

OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-231 — Myo-Inositol Complex Capsules** (amazon.de). Prepared for review.

1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is $P(0)=e^{-\lambda}$: ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	3 <= reach < 30 days AND throttled (price >= +8% and/or ad spend cut >= 50%)	Miss (voluntary)
4	Marketplace gap	Units = 0 on a day enclosed by sales, lambda >= 3/day	Lost (involuntary)

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = max(lambda - actual units, 0)** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

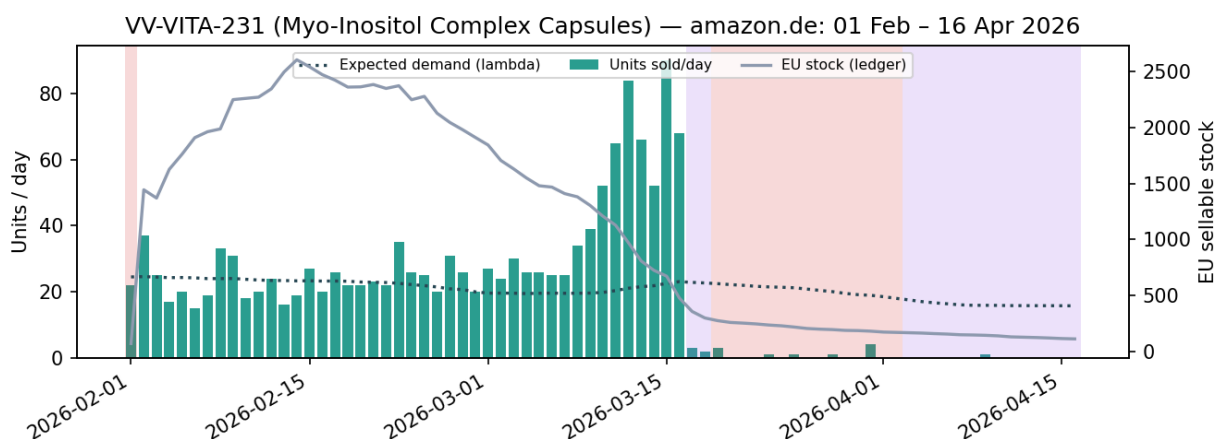
Lost / Miss revenue = missed units x avg price

Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-231 — Myo-Inositol Complex Capsules, amazon.de. Full year on this marketplace: **37 OOS days** (lost €21.713 revenue / €6.371 CM3) and **27 cooling-down days** (miss €10.727 revenue / €2.087 CM3).



Highlighted window 01 Feb 2026 – 16 Apr 2026: EU stock ranged from ~2.607 down to ~70 units and reach bottomed at ~2 day(s). In this window the model flagged **16 OOS day(s)** (red/orange shading) and **16 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
02-01	22	24	70	2	26.90	Critically low (<3d)	21	
02-02	37	25	1445	55	26.91	—		
02-17	26	23	2425	59	27.14	—		
03-04	26	19	1551	20	26.88	—		
03-17	3	23	357	4	27.15	Cooling down		116
03-18	2	23	299	3	26.90	Cooling down		119
03-19	3	22	275	3	26.90	Critically low (<3d)	111	
03-20	0	22	259	3	—	Critically low (<3d)	125	
03-21	0	22	253	3	—	Critically low (<3d)	122	
03-22	0	22	246	3	—	Critically low (<3d)	119	
03-23	1	21	236	3	26.90	Critically low (<3d)	112	
03-24	0	21	229	3	—	Critically low (<3d)	116	
03-25	1	21	218	3	26.90	Critically low (<3d)	109	
03-26	0	21	205	3	—	Critically low (<3d)	112	
03-27	0	20	199	3	—	Critically low (<3d)	109	
04-03	0	18	167	3	—	Cooling down		88
04-04	0	17	164	3	—	Cooling down		84
04-05	0	17	159	3	—	Cooling down		82
04-06	0	16	155	3	—	Cooling down		81
04-07	0	16	149	4	—	Cooling down		78
04-08	0	16	147	4	—	Cooling down		78

6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.